Customer Behavior Analysis Using E-Commerce Data

Data Exploration, SQL Analysis, and Business Insights

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Introduction

This project focuses on analyzing customer behavior using an e-commerce dataset. The dataset contains information about users, orders, products, and payments, which together represent the full customer purchasing journey. By studying this data, it is possible to understand how customers interact with the platform and identify patterns in their purchasing behavior.

Understanding customer behavior is important for businesses because it helps them make better decisions about marketing strategies, product offerings, and customer engagement. Data analysis allows organizations to identify trends, measure performance, and discover opportunities for improvement.

In this project, SQL will be used to explore the dataset and analyze relationships between different tables. The analysis aims to uncover useful insights such as purchasing patterns, product performance, and overall business trends. These insights can help businesses improve their strategies and enhance customer experience.

DATASET SELECTION

The dataset used in this project contains transactional and behavioral data from an e-commerce platform. It includes information about users, orders, products, and payments. This type of dataset is valuable because it captures multiple stages of the customer journey, from user registration to product purchase and payment completion. By analyzing these different components together, it becomes possible to understand how customers interact with the platform and how their actions contribute to revenue generation.

This dataset was selected because it enables meaningful analysis of customer behavior and purchasing patterns. It allows exploration of questions related to user engagement, purchase frequency, and product popularity. Additionally, the dataset structure supports relational analysis across multiple tables, which makes it suitable for SQL-based investigation and data modeling.

Understanding customer behavior through this data can provide insights into how businesses can improve customer experience, optimize product offerings, and increase conversion rates. These insights are important for organizations that rely on data-driven decision making to improve performance and growth.

DATA IMPORT AND CLEANING

The dataset was imported into a relational SQL database to allow structured querying and analysis. Each dataset file was converted into a corresponding table within the database. During this process, data types were verified to ensure consistency across the tables. For example, date columns were formatted correctly and numerical fields such as prices and quantities were stored in appropriate numeric formats.

Several data preparation steps were performed before analysis. Duplicate records were checked and removed where necessary to prevent inaccurate aggregation results. Missing values were identified and evaluated to determine whether they represented incomplete data or optional fields. In cases where missing values could affect analysis, appropriate filtering or validation steps were applied.

Primary and foreign key relationships were also examined to ensure that tables could be joined correctly during analysis. This step is critical when working with relational databases because accurate joins allow analysts to reconstruct the full lifecycle of a transaction across multiple tables.

These preparation steps ensure that the dataset is reliable and suitable for further exploration and analysis.

INITIAL DATA EXPLORATION

Initial exploration of the dataset was conducted to understand its structure and the relationships between tables. The primary entities within the dataset include users, products, orders, and payments. Each entity represents an important component of the e-commerce ecosystem.

Basic exploratory SQL queries were used to examine the size of each table, the number of users in the system, and the total number of orders and transactions recorded. These initial queries help confirm that the data has been loaded correctly and provide an overview of the scale of activity within the dataset.

Relationships between tables were also explored to understand how different events connect together. For example, a user may create multiple orders, and each order may contain multiple items. Payments are linked to orders, which allows tracking of completed transactions and revenue generation.

This early exploration phase provides the foundation for deeper analysis by identifying key fields, relationships, and metrics that will be important for answering business questions in later stages of the project.

ENTITY RELATIONSHIP DIAGRAM (ERD)

The ERD illustrates the relational structure of the e-commerce dataset and shows how the main entities are connected. The database consists of five key tables: Users, Orders, Products, Order_Items, and Payments. Primary keys uniquely identify records in each table, while foreign keys establish relationships between them. This structure allows analysis of the full customer journey, from user registration to order creation, product purchases, and payment completion.

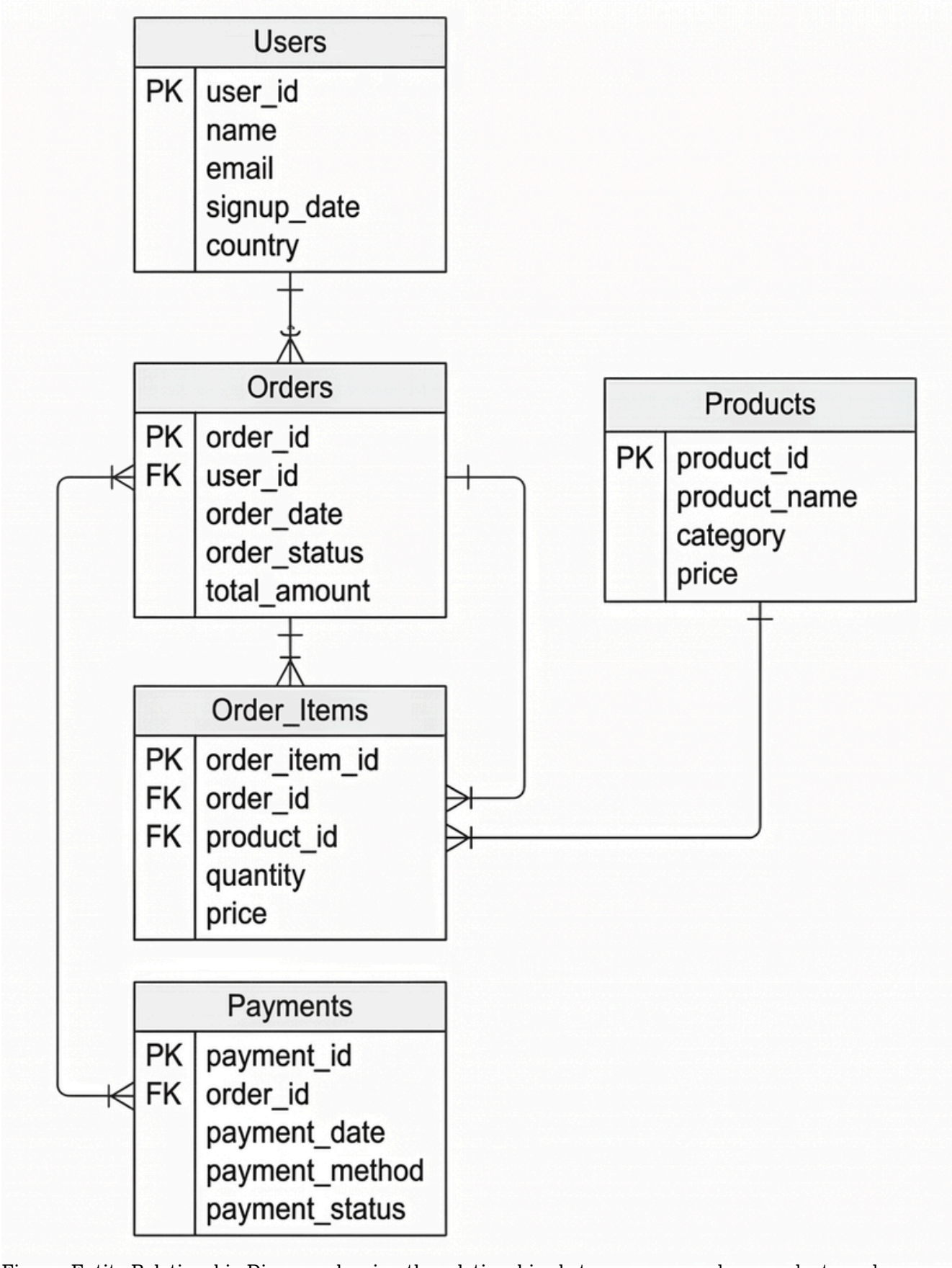

Figure: Entity Relationship Diagram showing the relationships between users, orders, products, order items, and payments in the e-commerce database.

PROJECT DESCRIPTION

This project focuses on analyzing customer behavior using e-commerce transaction data. The dataset includes information about users, orders, products, and payments, which together represent the full lifecycle of a customer purchase. By examining the relationships between these tables, the analysis aims to uncover patterns in purchasing behavior, product performance, and customer engagement. Understanding these patterns can help businesses make better decisions regarding product strategy, marketing efforts, and customer retention. Through SQL-based analysis and relational data exploration, this project seeks to identify meaningful insights about how users interact with the platform and how those interactions contribute to overall revenue generation.

KEY QUESTIONS

- How does user activity change after signup, and what percentage of users convert into paying customers?
- What is the average number of orders placed per user, and how does purchasing frequency vary across customers?
- Which products or product categories generate the highest revenue on the platform?
- How does customer purchasing behavior evolve over time, and what patterns can be observed in repeat purchases?
- What factors appear to influence order value and overall revenue generation?

HYPOTHESES

Based on typical patterns observed in e-commerce platforms, several initial assumptions can be made before conducting the analysis. First, it is expected that a relatively small group of highly active users contributes a significant portion of the total revenue. Second, certain product categories are likely to generate more sales than others due to differences in customer demand and popularity. Finally, repeat customers are expected to place orders more frequently and contribute more consistently to overall revenue compared to first-time buyers.

APPROACH

The analysis will begin by examining key fields such as user_id, order_id, product_id, order_date, and total_amount to understand how different entities in the dataset relate to each other. SQL queries will be used to join the Users, Orders, Products, Order_Items, and Payments tables in order to reconstruct the full customer journey from signup to purchase. Aggregation functions such as COUNT, SUM, and GROUP BY will be used to measure user activity, product performance, and revenue distribution. These metrics will help identify patterns in purchasing behavior and provide insights into how customer engagement contributes to overall revenue generation.